PRECALCULUS · UNIT 2: FUNCTIONS

Streaming Service Analyst

Use function families, transformations, compositions, and inverses to analyze a streaming business.

CP1

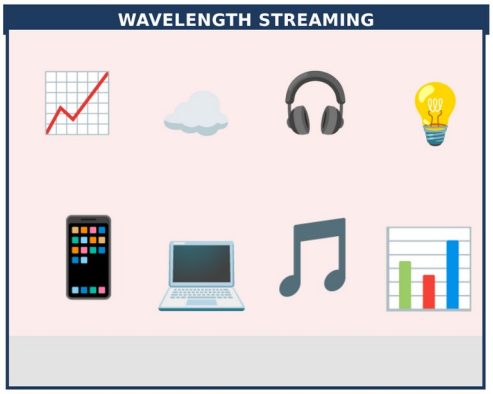
2–3 class periods

Name: _____ Date: _____ Period: _____

The Scenario

You are a data analyst at Wavelength, a new music streaming service. Leadership wants to know how pricing plans work, how fast subscribers are growing, and when the company will reach its goals.

You will use functions to model the business and present your findings.



Subscribers

$S(t) = 20{,}000(1.08)^t$

t in months

Plans

Basic $5 · Premium $11

Family $17

Learning Goals

- I can model with piecewise functions and transformations.
- I can compose functions and find inverses.
- I can use exponential and logarithmic functions to solve for time.

Project Timeline

When	What you will do
Day 1	Pricing Functions
Day 2	Transform the Data
Day 3	Logs & Goals
Final	Analyst Briefing + reflection

Part 1: Pricing Functions

Day 1 · Use the Business Data Cards.

Business Data Cards

Data Overage	Exchange	Tax
$2 per GB over 10 GB on the Basic plan	E(d) = 0.92d dollars → euros	T(x) = 1.0625x

Growth	Churn	Goal
8% per month from 20,000	Old app users fall 5% per month from 50,000	100,000 subscribers

1. Write the Basic plan cost as a piecewise function and graph it.

2. Write the cost with tax, T(C(g)), for g = 14.

3. Write E(T(x)) as a single function and find the euro price of the $11 Premium plan.

4. Find the inverse of E(T(x)). What does it convert?

Part 2: Transform the Data

Day 2 · $S(t) = 20{,}000(1.08)^t$.

5. A marketing campaign adds 5,000 subscribers at launch. Write the new function and describe the transformation.

6. The campaign starts 3 months late. Write $S(t - 3)$ and describe the shift.

7. Is $S(t)$ even, odd, or neither? Is it one-to-one?

8. Find the domain and range of $S(t)$ for $t \geq 0$.

Part 3: Logs & Goals

Day 3 · Use logarithms to solve for time.

9. Solve $20{,}000(1.08)^t = 100{,}000$ using logarithms.

10. Old app users: $U(t) = 50{,}000(0.95)^t$. When do they fall below 25,000?

11. When will new subscribers be twice the old app users?

Final Product: Analyst Briefing

Your final product must include:

- Piecewise pricing and graph
- Compositions and inverse with meanings
- Transformed subscriber models
- Logarithmic time calculations
- Analyst briefing

Reflection

R1. What does the composition E(T(x)) mean in words?

R2. Why do we use logarithms to solve for time in an exponential model?

Scoring Rubric

Criteria	4 – Exceeds	3 – Meets	2 – Approaching	1 – Beginning
Mathematical Accuracy	All calculations and solutions are correct; answers include units and are checked.	Nearly all work is correct; minor errors do not change conclusions.	Several errors; some conclusions are affected.	Many errors or missing work.
Reasoning & Justification	Clear, complete explanations connect every answer to the situation; uses precise vocabulary.	Explanations are mostly clear and connected to the situation.	Explanations are brief or partly unclear.	Little or no explanation.
Application & Design	Functions, transformations, compositions, inverses, and exponential models are built and interpreted correctly. Goes beyond the minimum with a creative, realistic choice.	Functions, transformations, compositions, inverses, and exponential models are built and interpreted correctly.	Design is incomplete or only partly uses the math from the unit.	Design is missing or unrelated to the unit math.
Communication & Presentation	Briefing includes models, graphs, calculations, and a recommendation. Polished, organized, and visually engaging.	Briefing includes models, graphs, calculations, and a recommendation. Organized and easy to follow.	Product is missing required parts or is hard to follow.	Product is mostly missing.